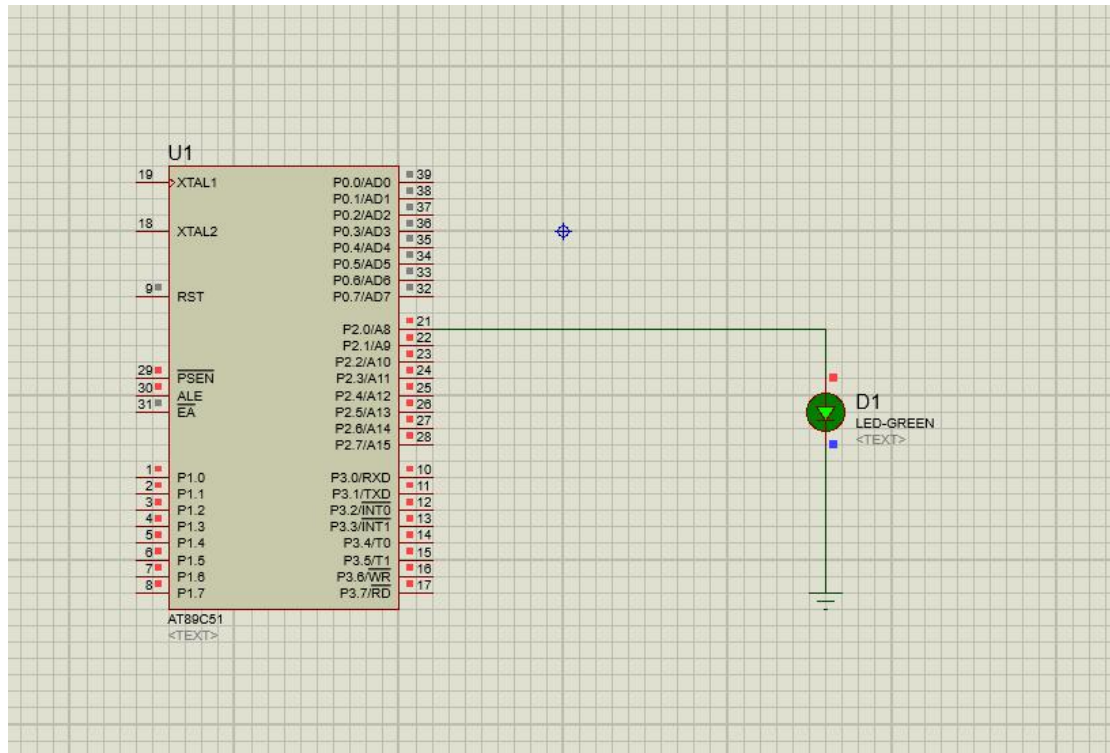
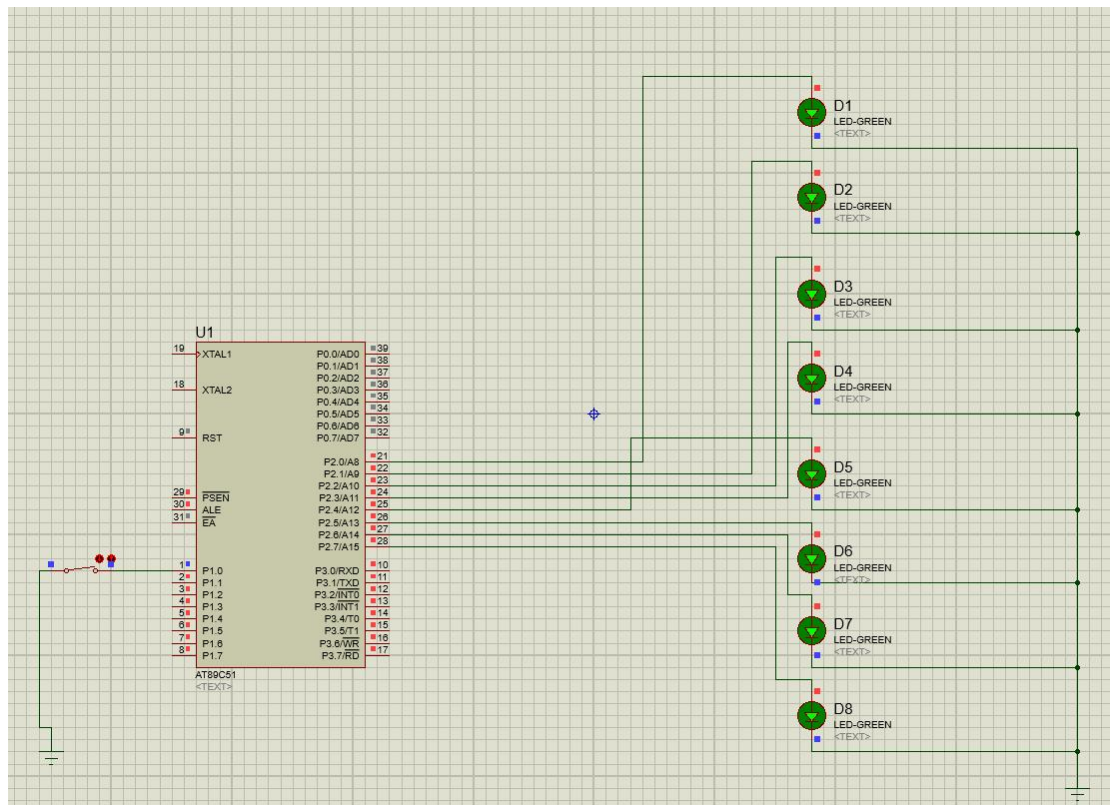


Led:



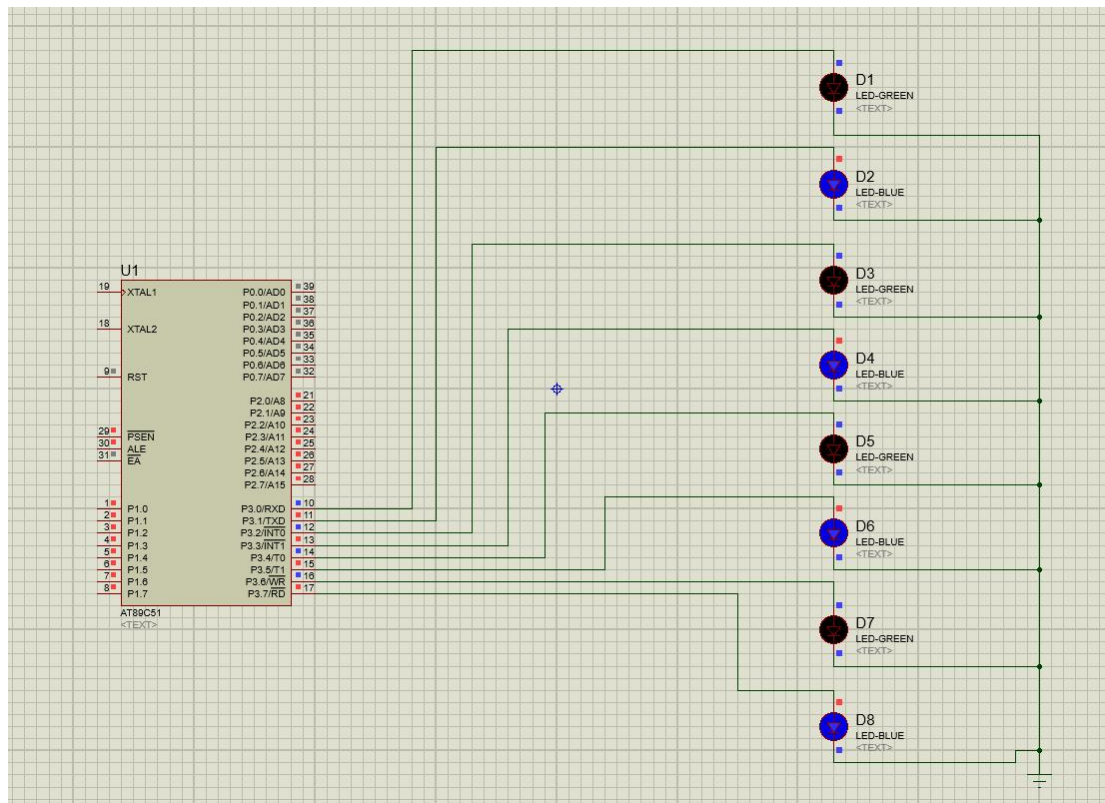
```
#include<reg51.h>
void delay(){
int m;
for(m=0;m<10000;m++);
}
sbit led=P2^0;
void main(){
led=0;
delay();
led=1;
delay();
}
```

8LED:



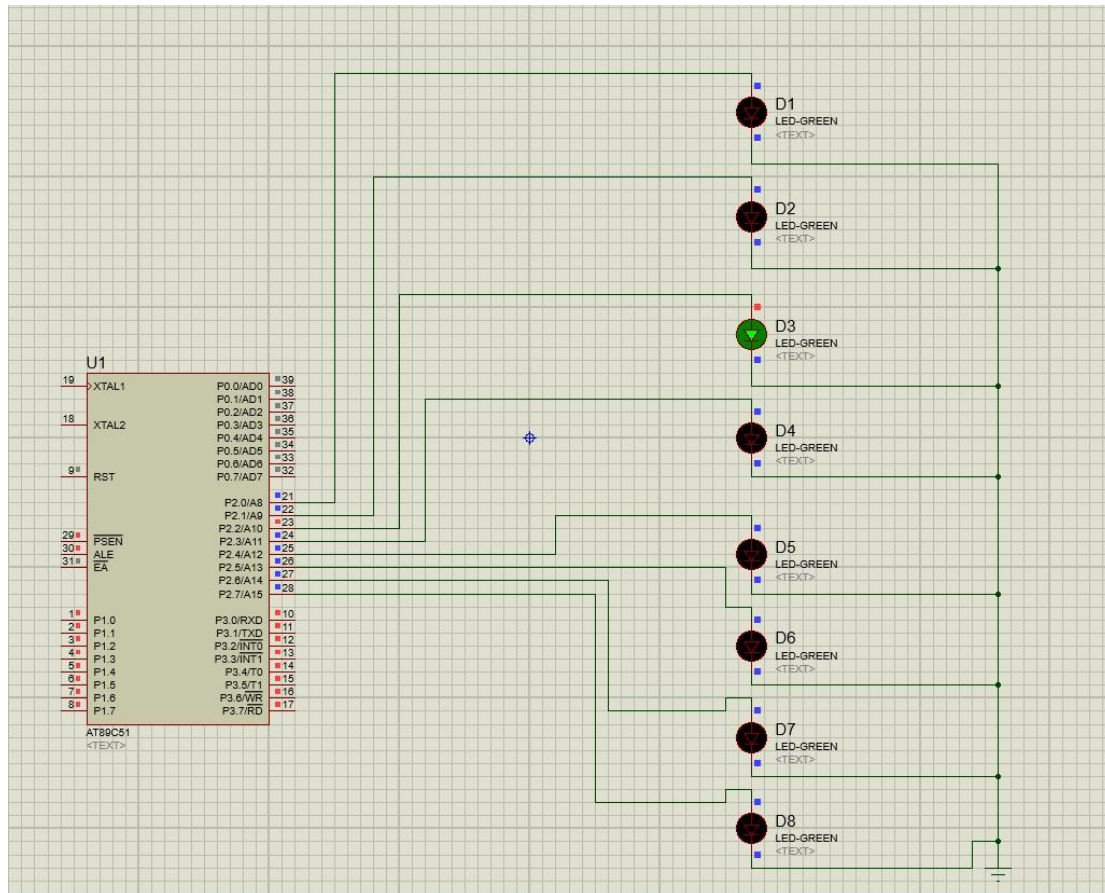
```
#include<reg51.h>
void delay(){
    int i;
    for(i=0;i<10000;i++);
}
sbit s=P1^0;
void main(){
    if(s==0){
        P2=0xFF;
        delay();
    }
    else{
        P2=0x00;
        delay();
    }
}
```

Task1 alter led:



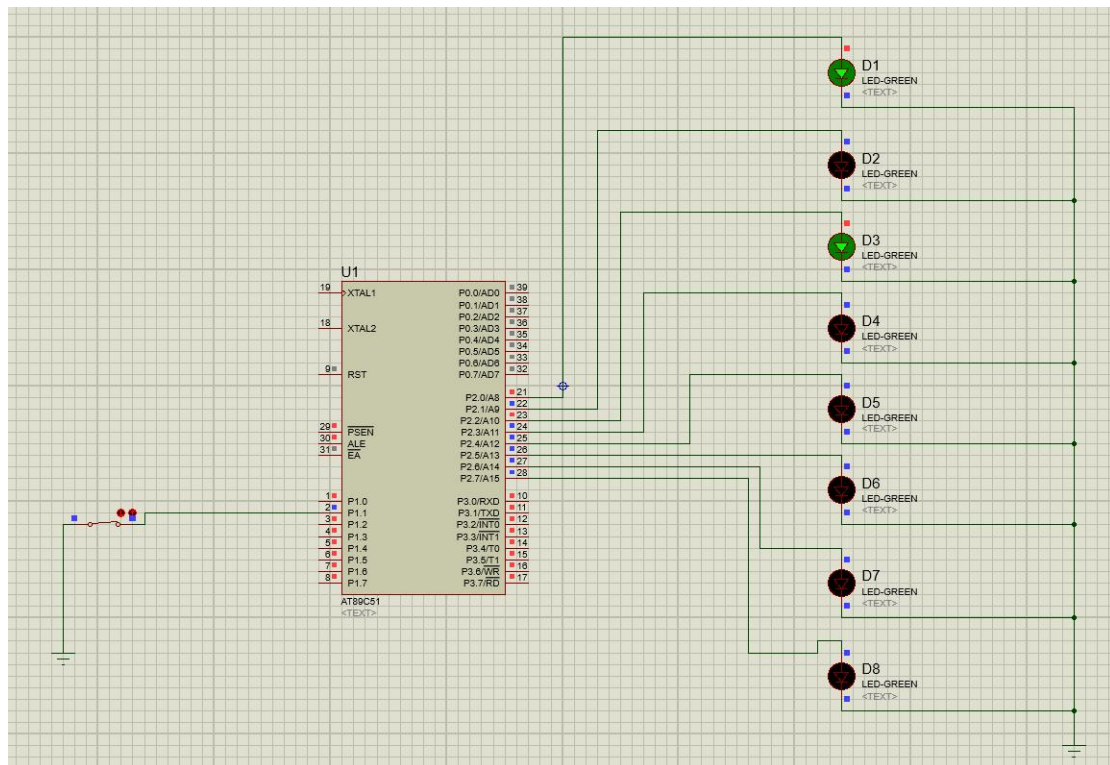
```
#include<reg51.h>
void delay(){
    unsigned int i,j;
    for(i=0;i<300;i++){
        for(j=0;j<1000;j++);
    }
}
void main(){
    while(1){
        P3=0x55;
        delay();
        P3=0xAA;
        delay();
    }
}
```

Task1 cyclic led(one LED ON at a time):



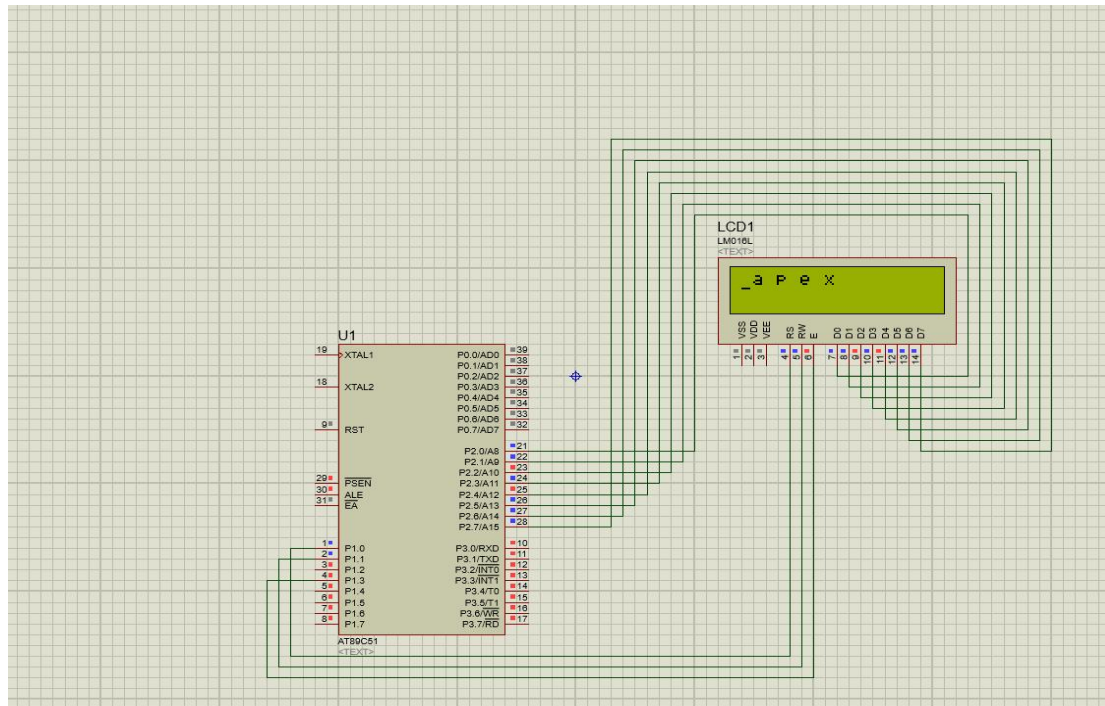
```
#include<reg51.h>
void delay(){
    unsigned int i,j;
    for(i=0;i<300;i++){
        for(j=0;j<1000;j++);
    }
}
void main(){
    unsigned char LED;
    while(1){
        for(LED=0;LED<8;LED++){
            P2=1<<LED;
            delay();
        }
    }
}
```

Task1 count led(switch press count Increase):



```
#include<reg51.h>
sbit s1=P1^1;
void delay(){
    unsigned int i,j;
    for(i=0;i<300;i++){
        for(j=0;j<1000;j++){
        }
    }
}
void main(){
    unsigned int c=0;
    while(1){
        if(s1==0){
            c++;
            P2=c;
            delay();
            while(s1==1);
            delay();
        }
    }
}
```

LCD:



```
#include<reg51.h>
void delay(){
int i;
for(i=0;i<15000;i++){
}
sbit RS=P1^0;
sbit RW=P1^1;
sbit EN=P1^3;
void cmd(int x){
P2=x;
RS=0;
RW=0;
EN=1;
delay();
EN=0;
}
void dat(int y){
P2=y;
RS=1;
RW=0;
EN=1;
delay();
EN=0;
}
void main(){
cmd(0x38);
cmd(0x80);
cmd(0x06);
cmd(0x0E);
cmd(0x14);
dat('a');
```

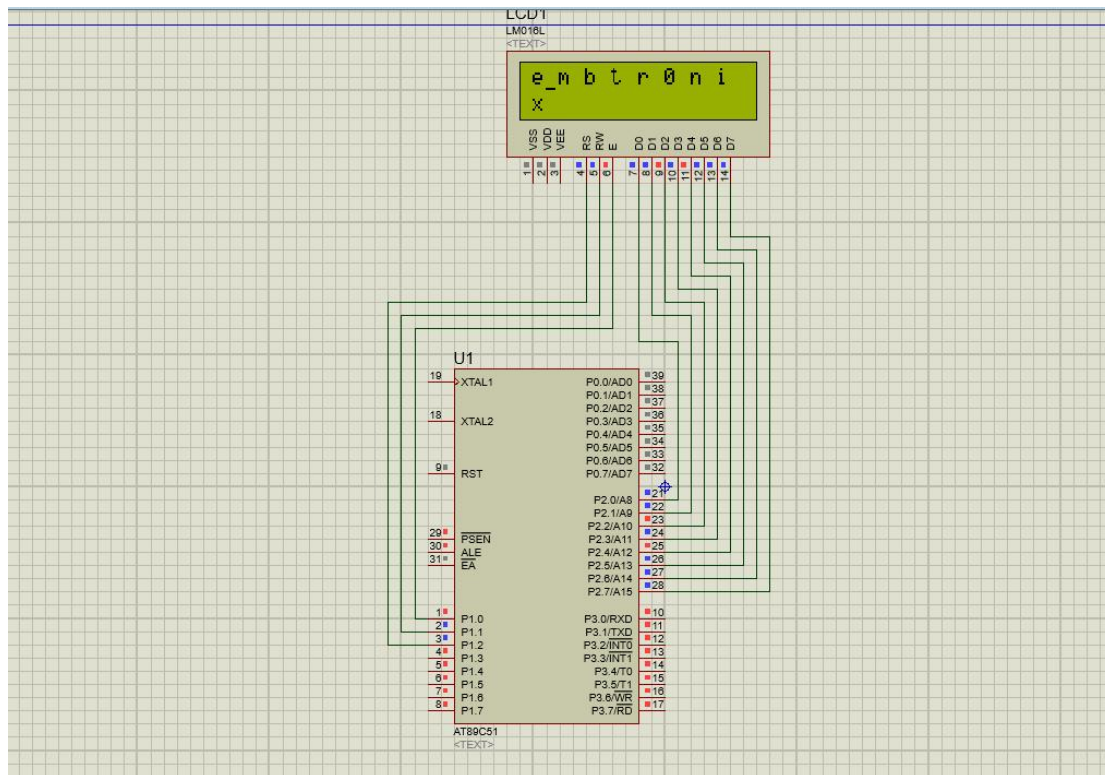


```

cmd(0x14);
dat('p');
cmd(0x14);
dat('e');
cmd(0x14);
dat('x');
}

```

Task2 LCD:



```

#include<reg51.h>
sbit RS=P1^2;
sbit RW=P1^1;
sbit EN=P1^0;
void delay(){
    unsigned int i,j;
    for(i=0;i<300;i++){
        for(j=0;j<1000;j++){
        }
    }
}
void cmd(unsigned char x){
    P2=x;
    RS=0;
    RW=0;
    EN=1;
    delay();
    EN=0;
}
void dat(unsigned char a){
    P2=a;
}

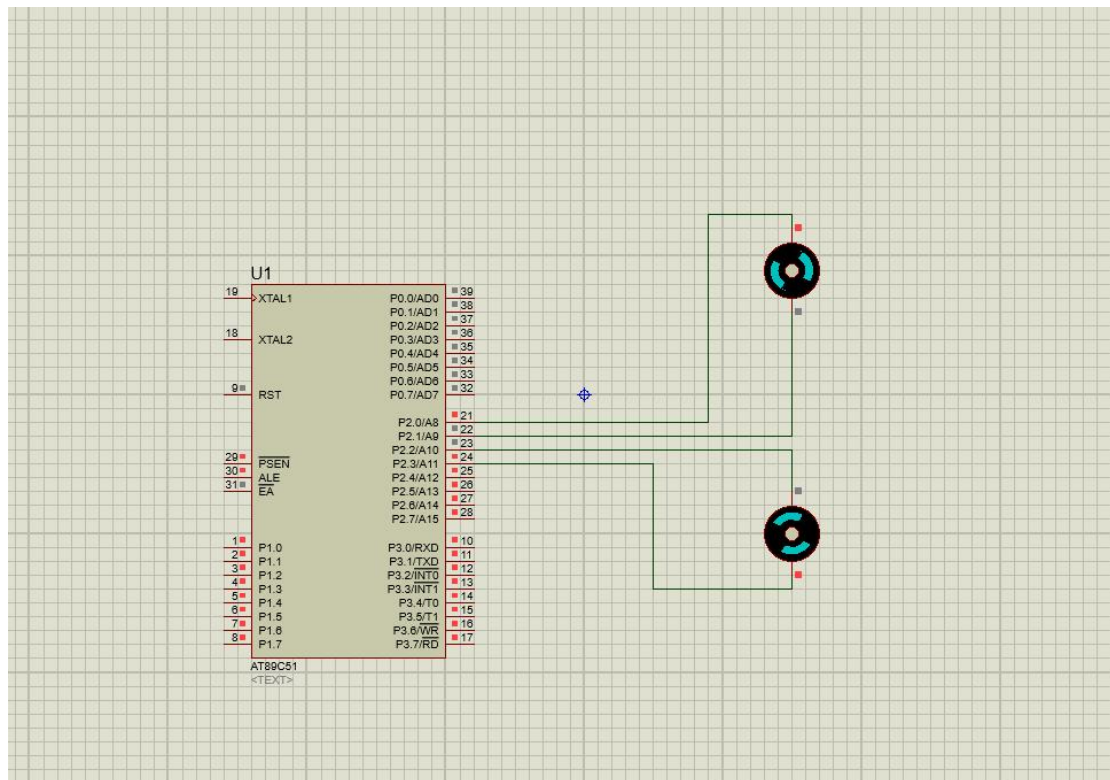
```

```
RS=1;
RW=0;
EN=1;
delay();
EN=0;
}
```

```
void main(){
while(1){
cmd(0x38);
cmd(0x06);
cmd(0x0E);
cmd(0x80);
dat('e');
        cmd(0x14);
dat('m');
        cmd(0x14);
dat('b');
        cmd(0x14);
dat('t');
        cmd(0x14);
dat('r');
        cmd(0x14);
dat('0');
        cmd(0x14);
dat('n');
        cmd(0x14);
dat('i');
        cmd(0xc0);
dat('x');

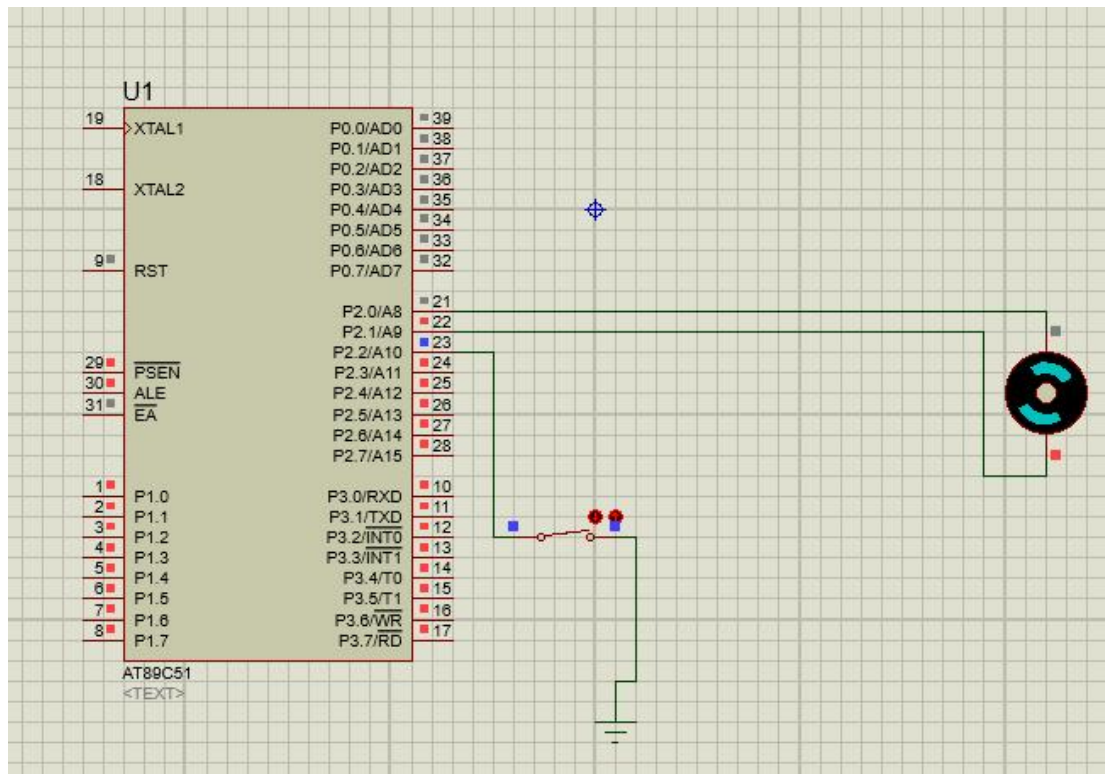
}
}
```


2motors:



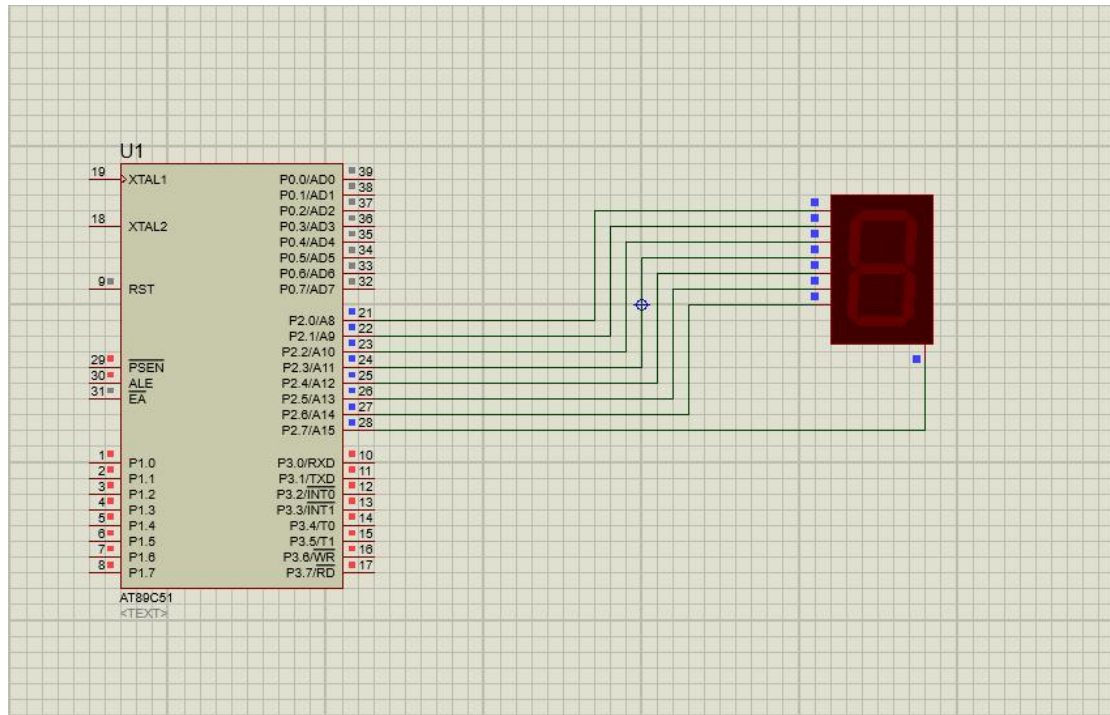
```
#include<reg51.h>
sbit m1=P2^0;
sbit m=P2^1;
sbit m2=P2^2;
sbit m4=P2^3;
void delay(){
    unsigned int i,j;
    for(i=0;i<500;i++){
        for(j=0;j<1000;j++);
    }
}
void main(){
    while(1){
        m1=1;
        m=0;//forward
        delay();
        m2=0;
        m4=1;//reverse
        delay();
        m1=0;m=0;
        m2=0;m4=0;//stop
        delay();
    }
}
```

MOTOR(switch):forward and reverse



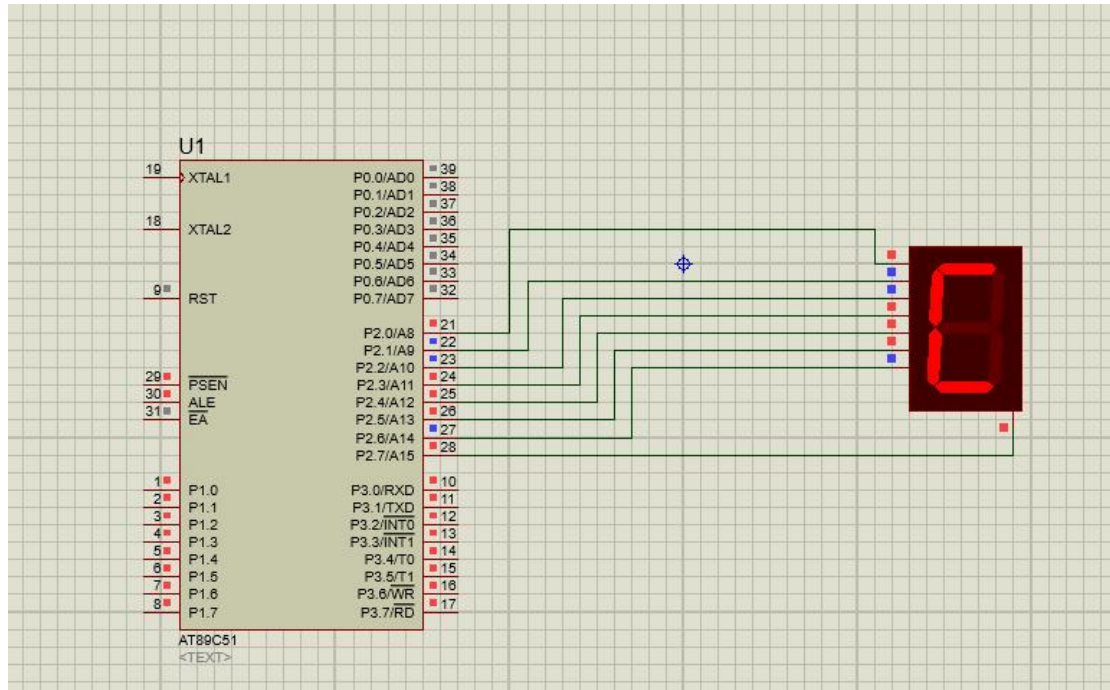
```
#include<reg51.h>
sbit m1=P2^0;
sbit m2=P2^1;
sbit s1=P2^2;
void delay(){
    unsigned int i,j;
    for(i=0;i<500;i++){
        for(j=0;j<1000;j++){
        }
    }
}
void main(){
    while(1){
        if(s1==1){
            m1=1;
            m2=0;
            delay();
        }
        else{
            m1=0;
            m2=1;
            dealy();
        }
    }
}
```

7segment:(0-9)



```
#include<reg51.h>
void delay(){
int n;
for(n=0;n<10000;n++);
}
void main(){
int a[16]={0x3F,0x06,0x5B,0x4F,0x66,0x6D,0x7D,0x07,0x7F,0x6F};
int m;
for(m=0;m<16;m++){
P2=a[m];
delay();
}
}
```

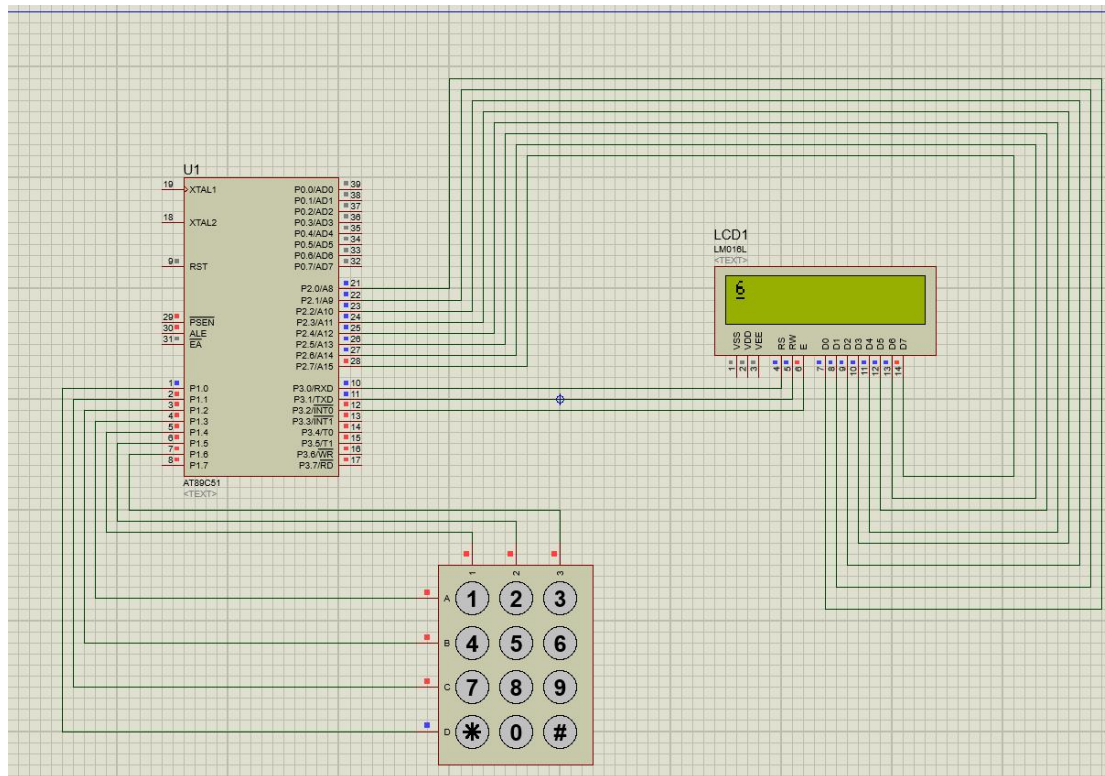
7segment(0-f):



```
#include<reg51.h>
void delay(){
int n;
    int i;
for(n=0;n<10000;n++){
    for(i=0;i<15000;i++){

}
void main(){
int a[18]={0x3F,0x06,0x5B,0x4F,0x66,0x6D,0x7D,0x07,0x7F,0x6F,0x77,0x7C,0x39,0x5E,0x79,0x71};
int m;
for(m=0;m<16;m++){
P2=a[m];
delay();
}
}
```

Keypad:



```
#include<reg51.h>
sbit rs=P3^0;
sbit rw=P3^1;
sbit en=P3^2;
sbit r1=P1^3;
sbit r2=P1^2;
sbit r3=P1^1;
sbit r4=P1^0;
sbit c1=P1^4;
sbit c2=P1^5;
sbit c3=P1^6;
void delay(){
    unsigned int i;
    for(i=0;i<1000;i++);
}
void cmd(int x){
    P2=x;
    rs=0;
    rw=0;
    en=1;
    delay();
    en=0;
}
void dat(int y){
    P2=y;
    rs=1;
    rw=0;
    en=1;
    delay();
}
```

```

en=0;
}
void main(){
cmd(0x38);
cmd(0x80);
cmd(0x06);
cmd(0x0E);
c1=c2=c3=1;
r1=0;r2=r3=r4=1;
if(c1==0){
dat('1');
delay();
}
if(c2==0){
dat('2');
delay();
}
if(c3==0){
dat('3');
delay();
}
r2=0;r1=r3=r4=1;
if(c1==0){
dat('4');
delay();
}
if(c2==0){
dat('5');
delay();
}
if(c3==0){
dat('6');
delay();
}
r3=0;r1=r2=r4=1;
if(c1==0){
dat('7');
delay();
}
if(c2==0){
dat('8');
delay();
}
if(c3==0){
dat('9');
delay();
}
r4=0;r1=r2=r3=1;
if(c1==0){
dat('*');
delay();
}
if(c2==0){
dat('0');
delay();
}
if(c3==0){

```

```
dat('#');  
delay();  
}  
}
```